

[Dealt with the issue of how to identify eichmuller space with abelian differentials, and measured foliations with 1-forms by fixing once and for all a covering of the topological torus]

1 Introduction

Ledrappier's 1999 paper *Distribution des orbites des réseaux sur le plan réel* [ref] proves an ergodic theorem for averages of $\mathrm{SL}(2, \mathbb{R})$ orbits of real-valued functions. The goals of this document are twofold (i) rewrite Ledrappier's paper in the language of Teichmüller theory, to better motivate and understand the result, and (ii) to explicate terse elements of Ledrappier's proof.

The following notation will be used throughout

- $\mathbb{T}^2$ will denote the (topological) torus
- $\mathbf{T}_1$ will denote the Teichmüller space of $\mathbb{T}^2$
- Γ will denote the mapping class group of $\mathbb{T}^2$
- $\mathcal{Q}^1$ will denote the space of unit quadratic differentials over $\mathbf{T}_1$
- $\mathcal{M}$ will denote the moduli space $\mathbf{T}_1/\Gamma$
- $\mathcal{MF}$ will denote the space of measured foliations on $\mathbb{T}^2$.
- Fix once and for all curves α, β with homology classes giving a basis for $\mathbb{T}^2$ and $i(\alpha, \beta) = 1$. Moreover, we may fix once and for all a covering map $\mathbb{R}^2 \rightarrow \mathbb{T}^2$, with (α, β) corresponding to the ordered lattice generators $((1, 0), (0, 1))$.

2 Teichmüller Theory for the Torus

Definition 1 *The Teichmüller space of the torus $\mathbf{T}_1$ will be defined as the set of pairs (S, φ) where S is a torus with a flat metric, and $\varphi : \mathbb{T}^2 \rightarrow S$ is an orientation preserving diffeomorphism, two pairs $(S_1, \varphi_1), (S_2, \varphi_2)$ will be identified if there is an isometry $I : S_1 \rightarrow S_2$, such that $I\varphi_1$ is homotopic to φ_2 . By the uniformization theorem we may equivalently consider Riemann surfaces and markings associated by homotopically trivial biholomorphism.*

In order to work with Teichmüller space, I will identify it with the space of abelian differentials of the form $dx + \tau dy$ for $\tau \in \mathbb{H}$, I will do this by first identifying $\mathbf{T}_1$ with representatives of the form $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, then pulling back the flat structure on the torus, by using the fixed covering on $\mathbb{T}$.

Start with a point $(S, \varphi) \in \mathbf{T}_1$, then we get an ordered pair of $\pi_1(S)$ generators from our marking $\varphi_*(\alpha)$ and $\varphi_*(\beta)$. Considering the covering map $\mathbb{C} \rightarrow S$, we can consider $A = 0 \cdot \varphi_*(\alpha), B = 0 \cdot \varphi_*(\beta)$ so that S is isometric to $\mathbb{C}/(A\mathbb{Z} + B\mathbb{Z})$. Such an identification with lattices is only defined up to linear isometry in $\mathbb{C}$, thus only up to rotation and dilation.

Conversely, we may start with a flat torus $S = \mathbb{C}/(A\mathbb{Z} + B\mathbb{Z})$, with $\pi_1(S)$ generated by α', β' with deck transformations giving A, B as above. There is a diffeomorphism $\varphi : \mathbb{T}^2 \rightarrow S$ with $\varphi_* : \alpha, \beta \mapsto \alpha', \beta'$, thus identifying $\mathbb{C}/(A\mathbb{Z} + B\mathbb{Z})$ with a point in Teichmüller space. Isometries of lattices homotopic to the identity must be linear transformations acting trivially on $H_1(S, \mathbb{Z})$, thus two marked lattices are identified with the same point in $\mathbf{T}_1$ exactly when they differ by a rotation and dilation.

We can choose a unique representative of each set of lattices up to rotation and dilation by multiplying by complex number A^{-1} , thus identifying our lattices with $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, for $\mathbb{Z} = A^{-1}B$. Since $\pi_1(S)$ acts by translation it is orientation preserving, thus the orientation on the universal cover $\mathbb{C}$ induces an orientation on S , thus our marking φ is orientation preserving when the pair of homology classes $\varphi_*[\alpha], \varphi_*[\beta]$ agrees with this induced orientation, this is exactly when $A^{-1}B \in \mathbb{H}$.¹

Given a point in $(S, \varphi) \in \mathbf{T}_1$, now identified with $(\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \varphi')$, there is a canonical abelian differential given by dz . Using our fixed covering $\mathbb{R}^2 \rightarrow \mathbb{T}^2$, we find that $\varphi'^*(dz) = dx + \tau dy$ our desired abelian differential. Moreover, this gives a natural identification $\mathbf{T}_1 \longleftrightarrow \mathbb{H}$.

Definition 2 *A quadratic differential on a Riemann surface S is a section of the symmetric square of the holomorphic cotangent bundle $T^{1,0} \otimes T^{1,0}$. For a quadratic differential q , its norm is defined as $\|q\| = \int_S |q|$, and the space of unit quadratic differentials on S is $\mathcal{Q}^1(S)$ is the set of quadratic differentials with norm 1. We define $\mathcal{Q}^1 = \{(S, \varphi, q) \mid (S, \varphi) \in \mathbf{T}_1\}$ and $q \in \mathcal{Q}^1(S)$.*

Consider a point $(S, \varphi) \in \mathbf{T}_1$, the coordinate charts on S are given by translation, thus for any two coordinates we have some constant so that $z_\alpha = z_\beta + c$. A holomorphic quadratic differential has a local expression $\phi_\alpha(z_\alpha)(dz_\alpha)^2$, and the transition laws for $T^{1,0} \otimes T^{1,0}$ are $\phi_\beta(z_\beta) = \phi_\alpha(z_\alpha) \left(\frac{dz_\alpha}{dz_\beta}\right)^2$, thus $\phi_\alpha(z_\alpha) = \phi_\beta(z_\beta)$ and since S is compact the maximum principle implies $\phi \in \mathbb{C}$ is a constant. Now for $\phi(dz)^2$ to be a unit quadratic differential, we require $|\phi| = \Im(\tau)^{-1}$ so that

$$1 = \int_S |q| = |\phi| \int_S |dz|^2 = |\phi| \text{Area}(S) = |\phi| \Im(\tau)$$

Identifying (S, φ) with its torus representative as above, and using our fixed covering for $\mathbb{T}^2$, we can associate each quadratic differential q to $\varphi^*(q) = \frac{e^{i\theta}}{\Im(\tau)}(dx + \tau dy)^2$, note that this gives a natural identification of $\mathcal{Q}^1 \longleftrightarrow T^1\mathbb{H}$ since each quadratic differential is uniquely determined by (θ, φ) . Since quadratic differentials on the torus are nonvanishing, we may take a locally well defined square root function, which allows us to associate 1-forms to q by considering $\Re(\sqrt{q}), \Im(\sqrt{q})$, where $\sqrt{q} = \sqrt{\phi}dz$. These naturally give rise to a foliation of the torus by considering their level sets, as well as a isotopy invariant measure for transverse curves by integrating their variation.

¹The identification of $\mathbf{T}_1$ with lattices follows roughly [ref FM]

Definition 3 A measured foliation of the torus is a foliation $\mathcal{F}$ of $\mathbb{T}^2$ into curves, equipped with a transverse measure μ on curves transverse to $\mathcal{F}$ satisfying μ is a locally finite borel measure on transverse curves and if I is an isotopy of curves such that $I([a, b], t)$ is transverse to $\mathcal{F}$ for all t , then $\mu(I([a, b], -))$ is constant. We identify two measured foliations if they are related by a homotopy of the torus.

Definition 4 We can define the space of projectivized measured foliations $\mathbb{P}\mathcal{MF}$ as $\mathcal{MF}/\mathbb{R}^\times$ where $\mathbb{R}^\times$ acts via $\lambda \cdot (\mathcal{F}, \mu) = (\mathcal{F}, \lambda\mu)$

Measured foliations on $\mathbb{T}^2$ may be associated with $(H^1(\mathbb{T}^2; \mathbb{R}) \setminus \{0\}/\pm) \cong \mathbb{RP}^1 \times \mathbb{R}_{>0}$. To see this, we take a basis η, ξ for $H^1(\mathbb{T}^2; \mathbb{R})$ satisfying

$$\int_{\alpha} \eta = 1 \quad \int_{\beta} \eta = 0 \quad \int_{\alpha} \xi = 0 \quad \int_{\beta} \xi = 1$$

Then we associate $\mathcal{F}$ to the unique (up to sign) cohomology class ω satisfying

$$i(\mathcal{F}, n\alpha + m\beta) = \left| \int_{n\alpha + m\beta} \omega \right|$$

writing ω in our basis, we associate $\mathcal{F}$ to $|b\eta + a\xi|$ for some $a, b \in \mathbb{R}$, similarly any nonzero cohomology class considered up to sign recovers a measured foliation $(\mathcal{F}, \mu)$ since by definition the leaves of our foliation are given by $\ker |\omega|$, with transverse measure $|\omega|$. Using our fixed covering $\mathbb{R}^2 \rightarrow \mathbb{T}^2$, we identify $(\eta, \xi) \longleftrightarrow (dx, dy)$, so that compatible with our notion of pullback for abelian and quadratic differentials each measured foliation can be written as $b dx + a dy$ for some $a, b \in \mathbb{R}$. We also fix an isomorphism of $\mathcal{MF} = (H^1(\mathbb{T}^2; \mathbb{R}) \setminus \{0\}/\pm)$ with $(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, this is done by taking $(a, b) \mapsto b\eta + a\xi$, which we identify with $b dx + a dy$, we will see later that this convention gives $\text{SL}(2, \mathbb{R})$ equivariance and thus is the correct convention to agree with Ledrappier's paper [ref]. Under this identification for $X \in \mathcal{MF}$ we find $[X] \in \mathbb{P}\mathcal{MF}$ is the projection to the $\mathbb{RP}^1$ coordinate.

Definition 5 The mapping class group of the torus is the set of orientation preserving diffeomorphisms $\mathbb{T}^2 \rightarrow \mathbb{T}^2$ with the relation $\gamma \sim \gamma'$ when they are homotopic. We denote the mapping class group of the torus as Γ .

The mapping class group acts naturally on $\mathbf{T}_1, \mathcal{MF}$, and $\mathcal{Q}^1$, these actions are respectively described as $\gamma : (S, \varphi) \mapsto (\gamma(S), \gamma\varphi)$ and the latter as $(\mathcal{F}, \mu) \mapsto (\gamma(\mathcal{F}), \gamma_*\mu)$, where for a transverse curve c , $\gamma_*\mu(c) = \mu(\gamma^{-1}(c))$ is the pushforward measure, and finally for a quadratic differential $q = \phi(dz)^2$ over a surface $S \in \mathbf{T}_1$ we simply pullback the differential $((S, \varphi), q) \mapsto (\gamma \cdot (S, \varphi), \phi(\gamma^* dz)^2)$. In order to work with measured foliations and relate them to unit quadratic differentials, we would like to use the cohomological description, thus we would like to compute explicitly the Γ action on $H^1(\mathbb{T}^2; \mathbb{R})$.

Since mapping classes on the torus are defined up to homotopy, a mapping class may be written as a map on $H_1(\mathbb{T}^2; \mathbb{Z})$, since these are invertible and

orientation preserving, the resulting matrix lies in $\mathrm{SL}(2, \mathbb{Z})$. Mapping class groups are in general generated by Dehn twists [ref Martelli 6.5.4]. A Dehn twist about a curve c is defined as taking a tubular neighborhood D with a (orientation preserving) diffeomorphism to $r : D \rightarrow S^1 \times [-1, 1]$, and a smooth bump function $\chi : [-1, 1] \mapsto [0, 1]$ which is zero near -1 and one near 1 , then defining $\gamma : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ via $x \mapsto r^{-1}(e^{2\pi i \chi} r(x))$ on D and identity elsewhere. This has the following pictorial description

[Insert image here]

From which we may identify the Dehn twist about α as the matrix $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$

and the dehn twist about β as $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, since these two matrices generate $\mathrm{SL}(2, \mathbb{Z})$ we have thus identified Γ with $\mathrm{SL}(2, \mathbb{Z})$.

Now, given a matrix $g \in \mathrm{SL}(2, \mathbb{Z})$, which is associated to $\gamma \in \Gamma$, then γ acts on $H^1(\mathbb{T}^2; \mathbb{R})$ via g^T , which is seen from the following duality relation, which is well defined since integration is defined up to homology classes by Cauchy's theorem.

$$\int_c \gamma^* \omega = \int_{S^1} c^* \gamma^* \omega = \int_{S^1} (\gamma c)^* \omega = \int_{\gamma_* c} \omega$$

Since we have associated both $\mathbf{T}_1$ and $\mathcal{MF}$ to cohomology classes, this definition suffices to describe both mapping class group actions since the $\mathbf{T}_1$ action takes $\varphi^* dz \mapsto \varphi^* \gamma^* dz$, agreement for measured foliations is a consequence of the duality relation. The pullback of cohomology classes agrees with taking the pullback of the quadratic differential by definition. [Should show this induces mobius transform on $\mathbb{H}$]

3 Teichmüller Metric and Dilations

[add some exposition here]

Definition 6 Given $q \in \mathcal{Q}^1$ lying above $(S, \varphi) \in \mathbf{T}_1$ we say that local coordinates (x, y) are natural coordinates when the horizontal and vertical foliations are given by $|dy|$ and $|dx|$ respectively.

Existence of natural coordinates on the torus is obvious, since given the lattice presentation of a torus, and $q = \phi(dz)^2$ we simply need to multiply the lattice coordinates by $\phi^{-\frac{1}{2}}$ since our differential has no zeroes we can take such a well defined branch of the square root function. This allows us to define Teichmüller mappings, which are mappings that shear a surface along a pair of foliations.

Definition 7 A map $h : S \rightarrow S'$ of Riemann surfaces is called Teichmüller when there are quadratic differentials q, q' on S, S' respectively such that for some $K \in \mathbb{R}_{>0}$ there are natural coordinates (x, y) near any point p and (x', y') near p' such that

$$h(x + iy) = \sqrt{K}x' + \frac{i}{\sqrt{K}}y'$$

Such a map is said to have dilation K .

By Teichmüller's theorem [ref FM] every homotopy class of maps has a unique Teichmüller mapping, given two $(S, \varphi), (S', \varphi') \in \mathbf{T}_1$ the homotopy class is always taken to be the change of markings $\varphi'\varphi^{-1}$. This allows us to define a distance function on Teichmüller space, in terms of the dilation of a Teichmüller mapping between the marked surfaces.

Definition 8 *The Teichmüller distance between two surfaces is defined as*

$$d_{\text{Teich}}(S, S') = \frac{1}{2} \log K$$

Where K is the dilation of a Teichmüller map in the homotopy class of the change of markings.

The Teichmüller distance gives a complete metric on $\mathbf{T}_1$ and the geodesics are in bijection with $\mathcal{Q}^1$, this is defined as taking $q \in \mathcal{Q}^1$ with natural coordinates (x, y) , then the Teichmüller geodesic $c(t)$ is given by taking the unit quadratic differential with natural coordinates $(t^{-1}x, ty)$. [ref FM]. Our goal is to understand the distribution of mapping class group orbits under the horocycle flow, which is related to the amount of shear a mapping class group element applies. Thus we will measure the size of a mapping class group element by the amount of shearing it applies to a surface

Definition 9 *For $\gamma \in \Gamma$ we define K_γ to be the dilation of the Teichmüller map $h : S \rightarrow \gamma(S)$ for $(S, \varphi) \in \mathbf{T}_1$. This is well defined since Γ acts on $(\mathbf{T}_1, d_{\text{Teich}})$ by isometry which is immediate since the Teichmüller map is now between coordinate charts composed with the same biholomorphism.*

Finally we define a section from $\mathcal{MF}$ to $\mathcal{Q}^1$, using our fixed covering of $\mathbb{T}^2$ as a basepoint, then for $\mathcal{F} \in \mathcal{MF}$ we will find that there is a unique $\mathcal{F}'$ with $[\mathcal{F}'] = [\mathcal{F}] \in \mathbb{P}\mathcal{MF}$ so that $\mathcal{F}'$ is the horizontal foliation for some $q \in \mathcal{Q}^1$ lying above dz . Thus we can recover a point in $\mathcal{Q}^1$ by travelling along the Teichmüller line associated to q until we get a quadratic differential with horizontal foliation $\mathcal{F}$. As discussed previously each quadratic differential is uniquely associated to one of the form $\frac{e^{i\theta}}{\Im \tau} (dx + \tau dy)^2$, thus fixing $\tau = i$ gives

$$\begin{aligned} \Im(\sqrt{q}) &= \sin(\theta/2)dx + \cos(\theta/2)dy \\ \Re(\sqrt{q}) &= \cos(\theta/2)dx - \sin(\theta/2)dy \end{aligned}$$

Thus for the measured foliation associated to $X = (a, b)$, we consider $q \in \mathcal{Q}^1$ with $\mathcal{F}$ with horizontal foliation $\|X\|^{-1}(bdx + ady)$ and vertical foliation $\|X\|^{-1}(adx - bdy)$, then shearing gives us a quadratic differential q with horizontal and vertical foliations given respectively by $bdx + ady$ and $\|X\|^{-2}(adx - bdy)$. So that the resulting $q \in \mathcal{Q}^1$ satisfies

$$\sqrt{q} = (a\|X\|^{-2} + ib) \left(dx + \frac{ai - \|X\|^{-2}b}{bi + \|X\|^{-2}a} dy \right)$$

Definition 10 *Written explicitly we get*

$$\begin{aligned}\Psi : \mathbb{RP}^1 \times \mathbb{R}_{>0} &\rightarrow \mathcal{Q}^1 \\ (a, b) &\mapsto (a\|X\|^{-2} + ib)^2 \left(dx + \frac{ai - \|X\|^{-2}b}{bi + \|X\|^{-2}a} dy \right)^2\end{aligned}$$

The last preliminary notion to understand is the Teichmüller horocycle flow, however it will be far more illustrative to understand this as the the unipotent component of the $\mathrm{SL}(2, \mathbb{R})$ action on $\mathcal{Q}^1$. Let $g \in \mathrm{SL}(2, \mathbb{R})$, then the action of g on $\mathcal{Q}^1$ is simply the matrix action on the natural coordinates [\[ref– defined as in Forni’s chapter 8 of dynamical systems handbook vol2\]](#). Let (x', y') be natural coordinates for quadratic differential q , then in order to keep conventions consistent, we take the action on natural coordinates as $g \begin{pmatrix} y' \\ x' \end{pmatrix}$, thus the action on cohomology classes is given by the pullback, so that the action is given explicitly as

$$\begin{pmatrix} \Im \sqrt{g \cdot q} \\ \Re \sqrt{g \cdot q} \end{pmatrix} = g^T \begin{pmatrix} \Im \sqrt{q} \\ \Re \sqrt{q} \end{pmatrix}$$

The Teichmüller unipotent flow is the action of the unipotent subgroup $\{h_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \mid s \in \mathbb{R}\}$

4 Statement of Theorem

Let $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, and suppose that $[X] \in \mathbb{RP}^1$ has irrational slope, then

5 Proof of Theorem

[\[Possibly include a section for topological considerations?\]](#)

6 References